Shannon Energy

Sales & Retention Analytics

Power BI demo project - end-to-end build

26,000 acquisitions · 17,000 churn events · 5 sales channels · 2023-2026

Raw CSV → cleaning in Power Query → star schema → 29 DAX measures → report

B2C channels only - B2B contracts analysed separately

Marina Kulman

Synthetic data. Built to demonstrate method.

The view a sales director opens in the morning

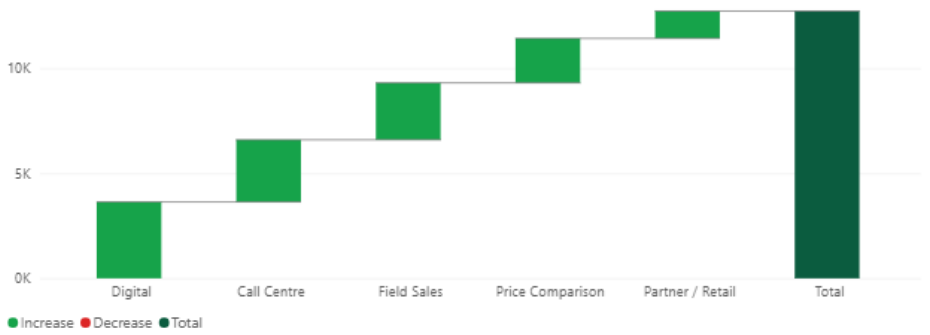
One page: volume, cost, plan vs actual, churn - filtered by segment and region.

Sales Performance — All segments · All regions

Closing Base	Net Adds	Acquisitions vs Target %	Monthly Churn Rate	CPA	Revenue Rolling 12M
18K	9K	-4.2%	2.9%	€ 112.30	€ 158M

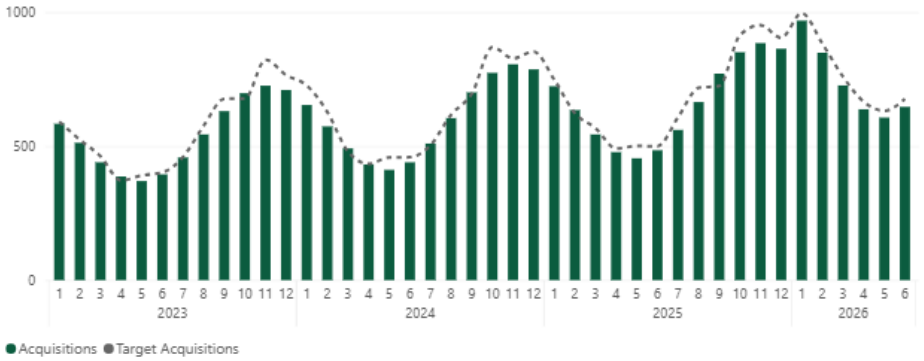
Net customer growth by channel

Net Adds by ChannelName



Acquisitions against plan

Acquisitions and Target Acquisitions by Year and MonthNumber



Plan delivery by channel

ChannelName	2024-05	2024-06	2024-07	2024-08	2024-09	2024-10	2024-11	2024-12	2025-01	2025-02	2025-03	2025-04	2025-05	2025-06	2025-07	2025-08	2025-09	2025-10	2025-11	2025-12	2026-01	2026-02	2026-03	2026-04	2026-05	2026-06	Total
Broker	-9.8%	2.6%	-4.7%	5.1%	-14.3%	-6.3%	0.0%	-13.5%	-8.3%	5.6%	-11.8%	13.5%	-11.4%	-4.9%	-13.6%	-14.9%	-13.8%	-4.2%	-4.7%	-15.3%	-12.8%	10.7%	1.8%	-2.5%	-17.8%	0.0%	**-6.6%**
Call Centre	-11.0%	-13.1%	-13.7%	0.9%	-0.8%	-13.7%	-10.8%	-11.1%	-5.3%	8.7%	4.7%	-1.2%	-9.6%	-3.1%	-14.2%	-2.2%	12.4%	-9.4%	-13.8%	-1.9%	-1.6%	9.2%	-15.0%	-9.0%	-11.0%	-14.0%	**-6.1%**
Digital	-14.1%	2.1%	9.5%	-6.0%	-1.2%	-15.2%	15.1%	-15.0%	7.1%	1.3%	-14.6%	8.7%	-9.4%	6.4%	-12.9%	-11.7%	4.8%	-9.5%	-4.2%	-1.5%	1.5%	-14.8%	2.8%	-14.9%	-0.7%	11.2%	**-3.8%**
Direct B2B	-14.3%	6.3%	-8.3%	-10.0%	10.0%	-8.7%	-16.7%	5.3%	-13.3%	0.0%	6.7%	11.1%	-8.3%	-12.5%	6.3%	0.0%	-4.3%	-13.3%	-10.0%	9.5%	-4.0%	-11.4%	-5.9%	-4.3%	5.6%	-8.0%	**-4.6%**
Field Sales	-12.9%	-9.7%	16.1%	9.0%	8.5%	0.0%	14.4%	0.8%	7.6%	10.7%	-3.7%	-16.1%	-11.6%	-16.1%	6.1%	-8.3%	6.2%	3.8%	1.5%	-0.9%	6.4%	13.2%	6.3%	-6.3%	-5.3%	-5.2%	**0.9%**
Partner / Retail	-12.2%	0.0%	-4.5%	2.2%	-17.2%	9.8%	-1.4%	-4.1%	-15.2%	-7.1%	-8.8%	2.6%	-8.5%	-7.1%	15.4%	-1.9%	10.0%	5.2%	-3.2%	-14.1%	-1.1%	0.0%	-10.3%	13.7%	14.6%	-12.7%	**-3.1%**
Price Comparison	-2.7%	-4.9%	1.6%	-7.6%	13.6%	-18.0%	-14.0%	-2.9%	-7.5%	-3.4%	4.2%	-10.7%	-6.7%	3.4%	-8.8%	-5.2%	10.4%	-10.6%	-10.9%	-4.5%	-9.0%	-11.7%	-6.6%	7.6%	-2.4%	-6.5%	**-5.5%**
Total	**-10.0%**	**-4.1%**	**1.2%**	**-2.1%**	**0.7%**	**-11.0%**	**-2.7%**	**-7.7%**	**-3.2%**	**1.9%**	**-4.1%**	**-2.6%**	**-9.2%**	**-3.0%**	**-7.7%**	**-7.4%**	**6.3%**	**-7.0%**	**-7.0%**	**-4.5%**	**-2.8%**	**-3.7%**	**-4.2%**	**-4.2%**	**-4.0%**	**-4.4%**	**-4.2%**

Every card is a DAX measure, not a hard-coded number.

The cheapest channel to acquire is the most expensive to keep a customer from

B2C channels: acquisitions, CPA, 12-month retention and the cost of each customer who stays.

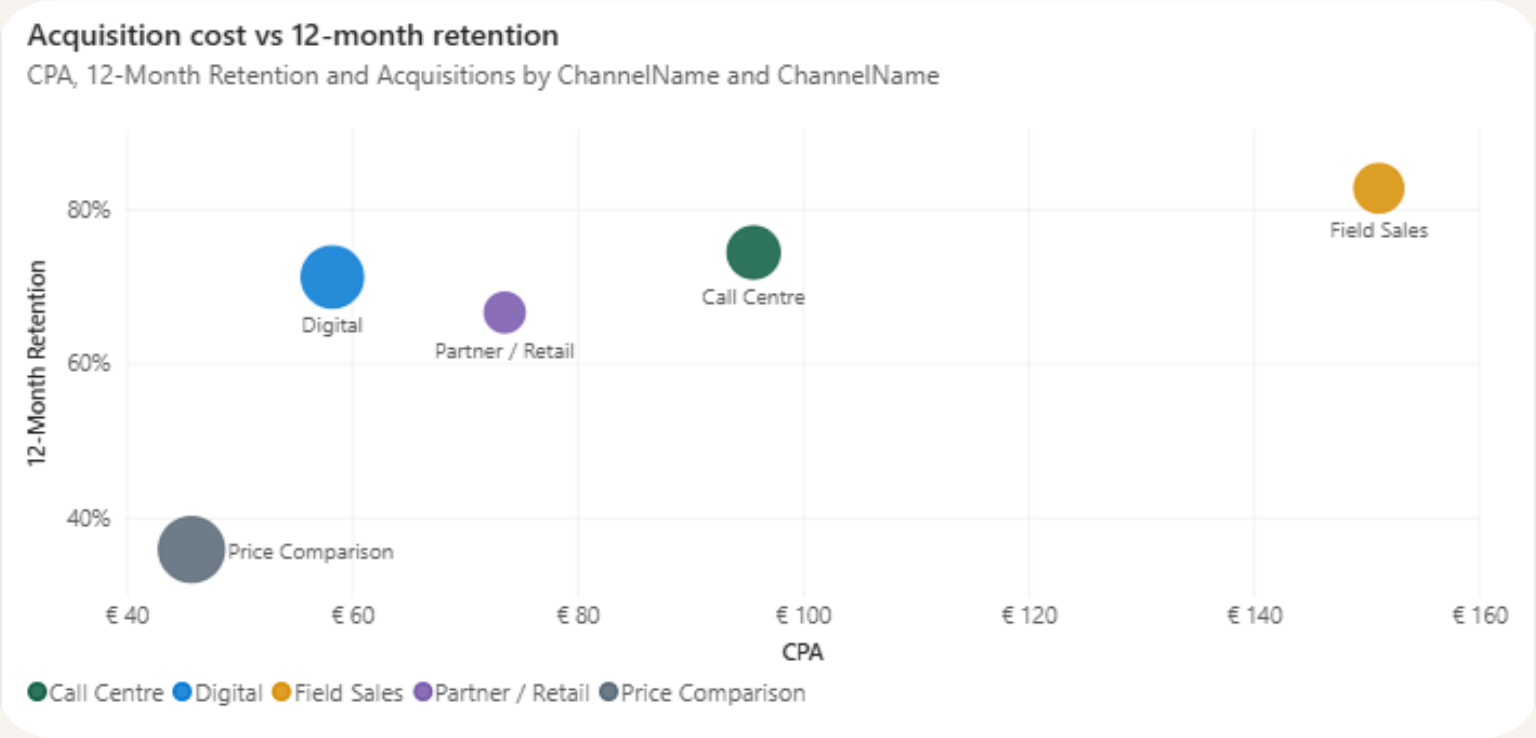
B2C channel economics

ChannelName	Acquisitions	CPA	12-Month Retention	Cost per Retained Customer
Price Comparison	6,619	€ 46	35.9%	€ 128
Digital	5,839	€ 58	71.2%	€ 82
Partner / Retail	2,279	€ 74	66.6%	€ 110
Call Centre	4,103	€ 96	74.4%	€ 129
Field Sales	3,548	€ 151	82.7%	€ 183
Total	**22,388**	**€ 78**	**62.6%**	**€ 124**

Price Comparison: CPA €46 - the lowest. Retention 35.9% - the lowest.

Low cost, low loyalty - Price Comparison sits alone

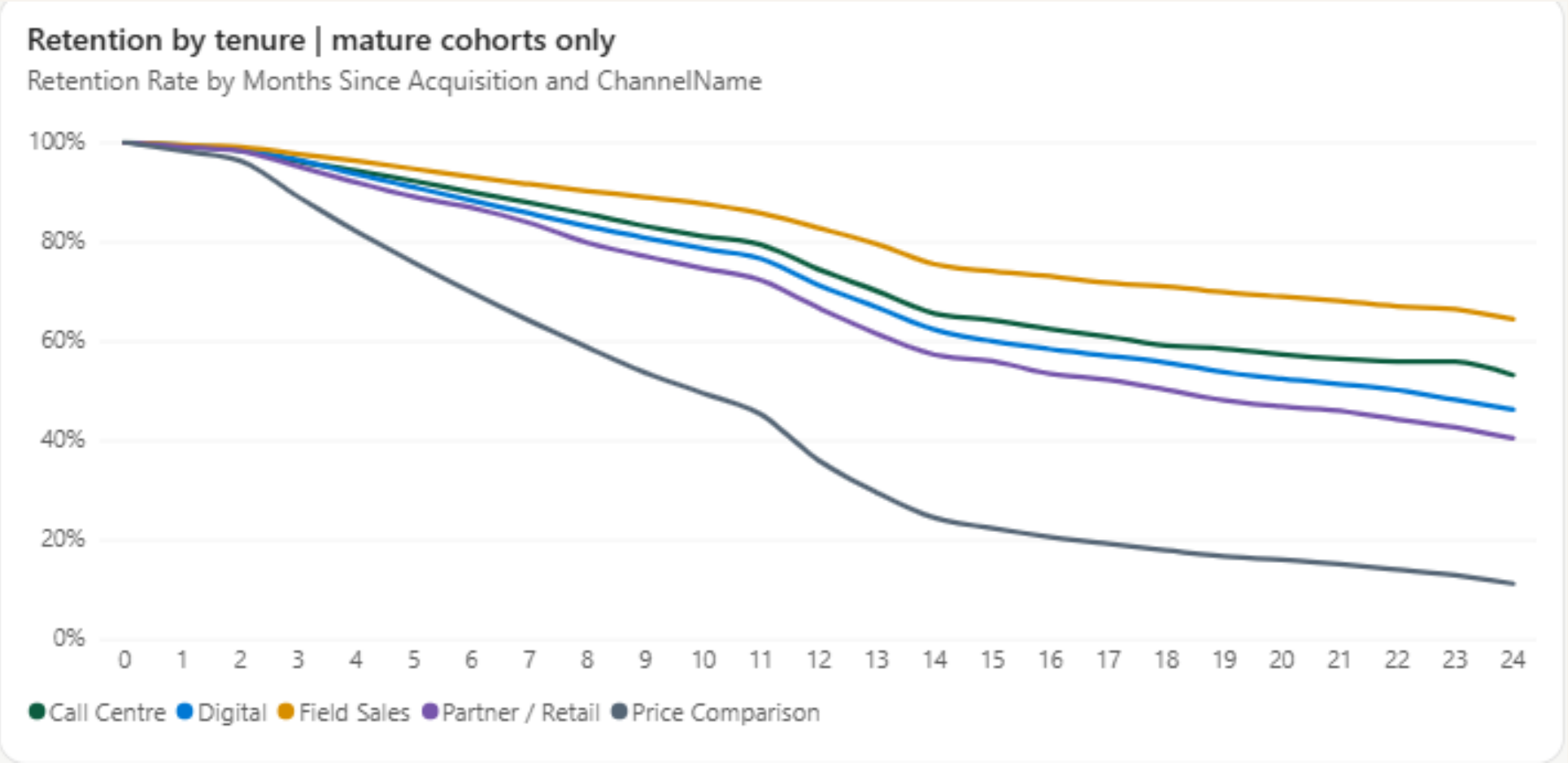
CPA against 12-month retention. Bubble size = acquisitions.



Cost per retained customer turns €46 into €128. Field Sales turns €151 into €183.

Price Comparison loses half its customers within eleven months

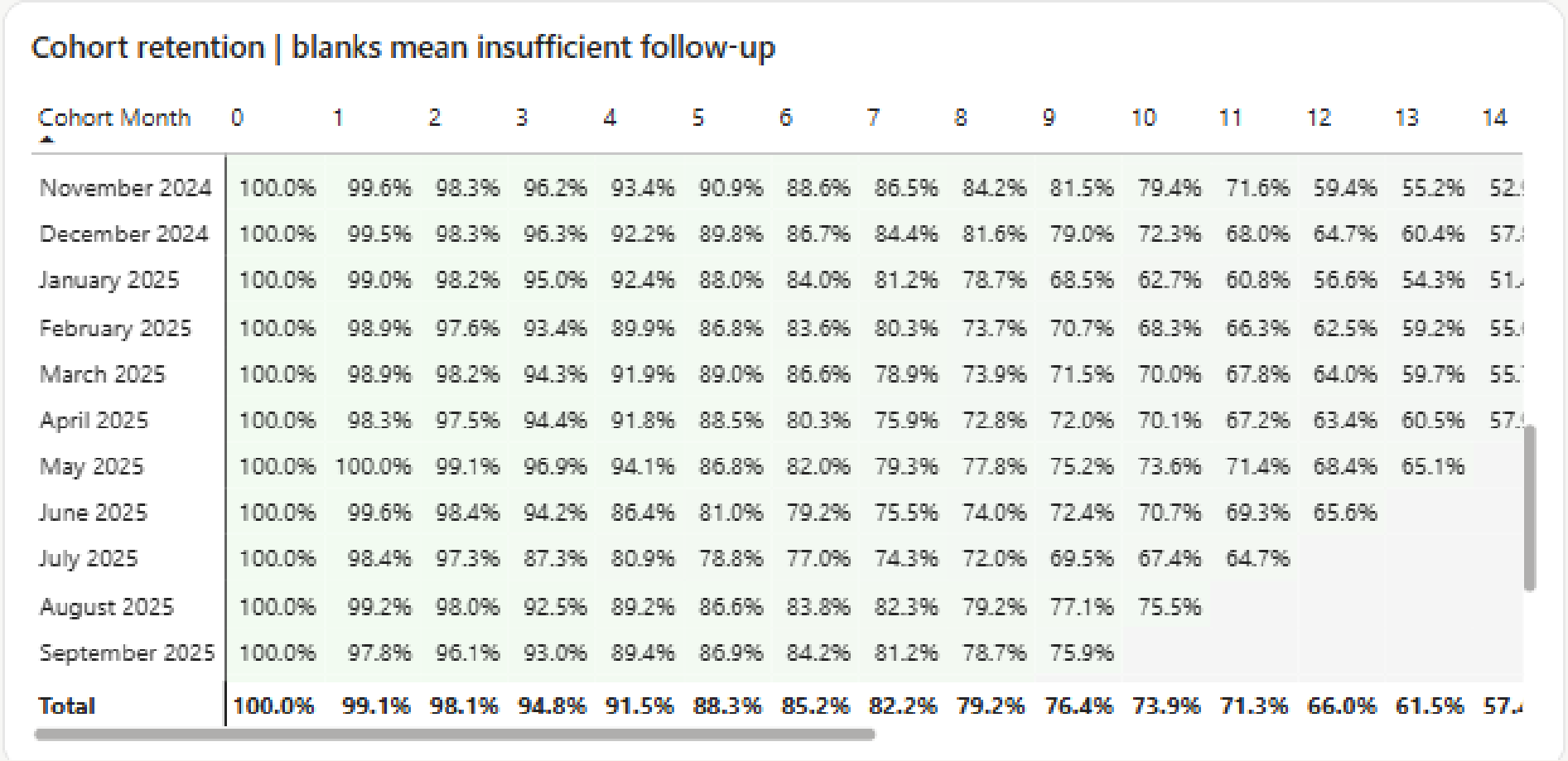
Retention by months since acquisition. The gap opens around month 12, where twelve-month contracts expire - a coincidence the data cannot explain.



Cohorts too young to have reached month N are excluded, not counted as retained.

Blanks are cohorts too young to measure

Cohort matrix: acquisition month * months since acquisition.



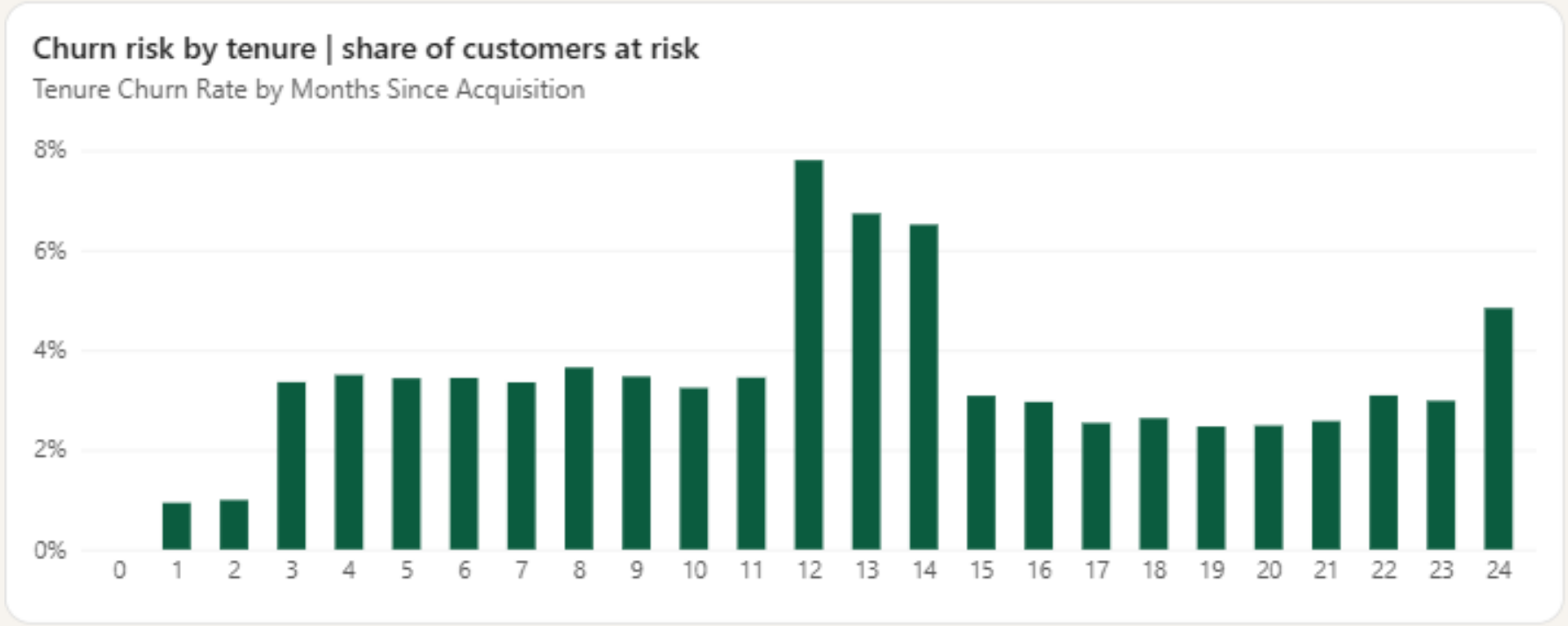
Cohort retention | blanks mean insufficient follow-up

Cohort Month	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
November 2024	100.0%	99.6%	98.3%	96.2%	93.4%	90.9%	88.6%	86.5%	84.2%	81.5%	79.4%	71.6%	59.4%	55.2%	52.
December 2024	100.0%	99.5%	98.3%	96.3%	92.2%	89.8%	86.7%	84.4%	81.6%	79.0%	72.3%	68.0%	64.7%	60.4%	57.
January 2025	100.0%	99.0%	98.2%	95.0%	92.4%	88.0%	84.0%	81.2%	78.7%	68.5%	62.7%	60.8%	56.6%	54.3%	51.
February 2025	100.0%	98.9%	97.6%	93.4%	89.9%	86.8%	83.6%	80.3%	73.7%	70.7%	68.3%	66.3%	62.5%	59.2%	55.
March 2025	100.0%	98.9%	98.2%	94.3%	91.9%	89.0%	86.6%	78.9%	73.9%	71.5%	70.0%	67.8%	64.0%	59.7%	55.
April 2025	100.0%	98.3%	97.5%	94.4%	91.8%	88.5%	80.3%	75.9%	72.8%	72.0%	70.1%	67.2%	63.4%	60.5%	57.
May 2025	100.0%	100.0%	99.1%	96.9%	94.1%	86.8%	82.0%	79.3%	77.8%	75.2%	73.6%	71.4%	68.4%	65.1%	
June 2025	100.0%	99.6%	98.4%	94.2%	86.4%	81.0%	79.2%	75.5%	74.0%	72.4%	70.7%	69.3%	65.6%		
July 2025	100.0%	98.4%	97.3%	87.3%	80.9%	78.8%	77.0%	74.3%	72.0%	69.5%	67.4%	64.7%			
August 2025	100.0%	99.2%	98.0%	92.5%	89.2%	86.6%	83.8%	82.3%	79.2%	77.1%	75.5%				
September 2025	100.0%	97.8%	96.1%	93.0%	89.4%	86.9%	84.2%	81.2%	78.7%	75.9%					
Total	**100.0%**	**99.1%**	**98.1%**	**94.8%**	**91.5%**	**88.3%**	**85.2%**	**82.2%**	**79.2%**	**76.4%**	**73.9%**	**71.3%**	**66.0%**	**61.5%**	**57.**

A blank is an honest gap in the data. Filling it with 100% is the classic error.

Churn more than doubles at month 12

Churned customers by tenure. The spike coincides with the end of twelve-month contracts - the data contains no renewal event or price, so this remains a hypothesis.



Whatever the cause, the timing is predictable - which makes it actionable before it happens.

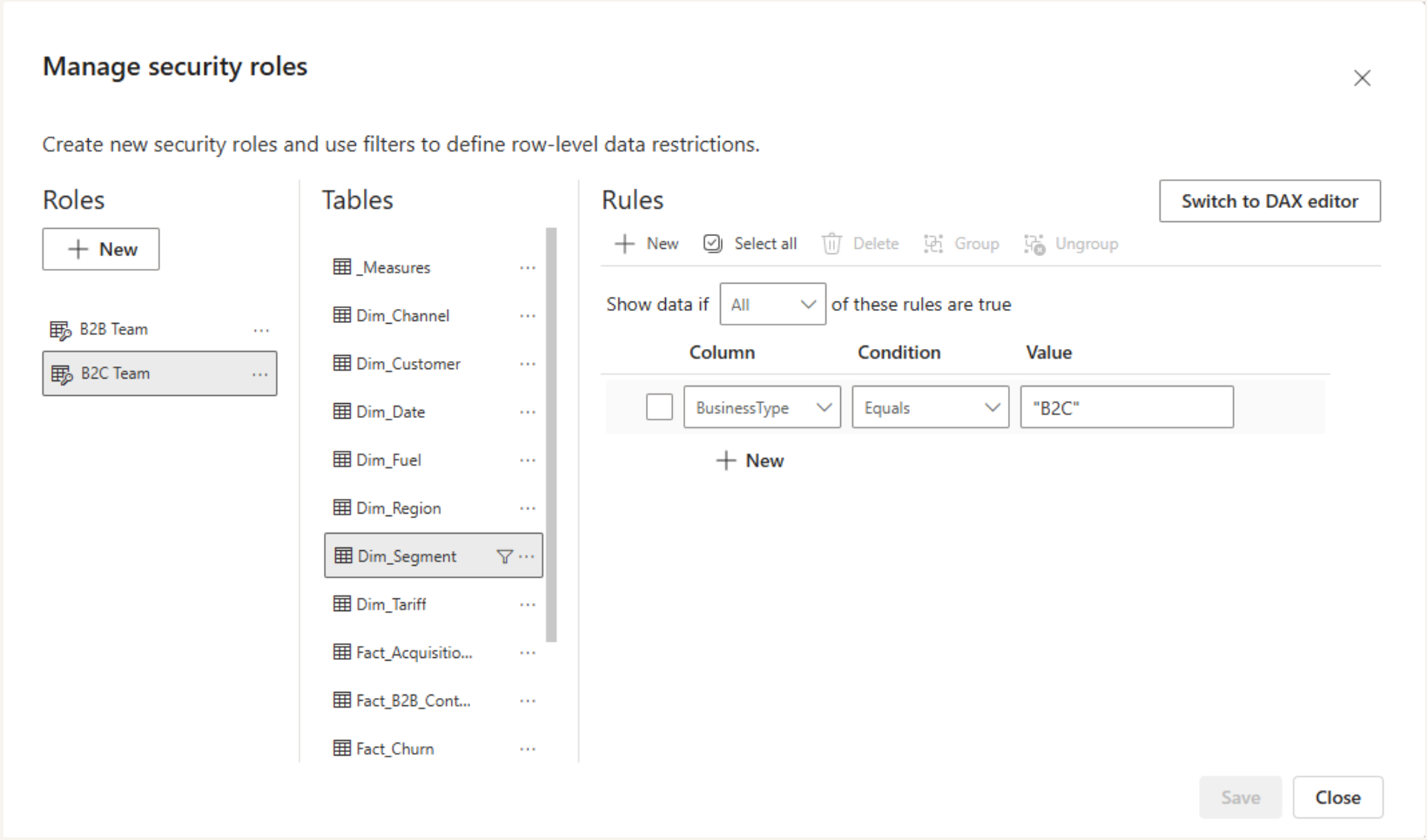
Cleaning and security are part of the build

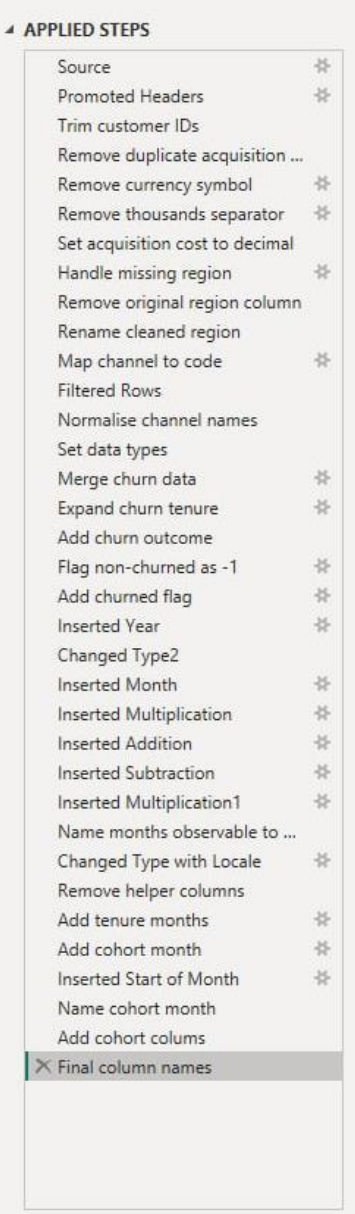
Applied Steps in Power Query · Row-Level Security roles.

Named steps, not "Changed Type1". Anyone can reopen this file and follow it.

Every transformation step, named ->

Twenty spellings of the channel, currency as text, mixed date formats





Two security roles

The B2C and B2B teams don't see each other's numbers

What the numbers say

€46	Price Comparison CPA - the cheapest channel to buy
35.9%	its 12-month retention - the worst of the five
€128	cost per retained customer, once churn is priced in
€82	what Digital pays for the same result

What this data cannot tell us

No cost of goods - margin and profitability cannot be calculated

No renewal events or prices - the month-12 pattern is a hypothesis, not a proven cause

No competitor pricing - switching reasons are self-reported

40% of churn events have no destination recorded

Synthetic data, fixed cut-off 30 June 2026

What to do about it

Move acquisition budget from Price Comparison to Digital
Digital delivers a retained customer for €82 against €128 - 36% less

Intervene at month 10, not month 13
Churn risk more than doubles at month 12 and the timing is predictable

Treat Price Comparison as a volume channel, not a growth channel
It brings the most customers and keeps the fewest

Capture renewal offers and outcomes
Without them the month-12 pattern stays a hypothesis